



Game Development Business

Common Game Team Roles[Careers]

http://creativeskillset.org/creative_industries/games/job_roles

Job Role	Job Description
Animator	responsible for the portrayal of movement and behaviour animations within a game (Walk , run, jump,...)
Game Designer	Designs what a game consists of and how it plays (GamePlay) , defining all the core elements (Characters , Environment, objects,...)
Audio Engineer	creates the soundtrack for a game, including music , sound effects, character voices and spoken instructions
Game Programmer	Game Programmers design and write the computer code that runs and controls a game
Lead Programmer	lead the programming team responsible for creating all the game's computer code
External Producer	External Producers ensure the successful delivery of a game, while working externally from the game development team
Games Artist	create the visual elements of a game, such as characters, scenery, objects, vehicles, surface textures and clothing
Level Editor	define and create interactive architecture for a level of a game, including the landscape , buildings and objects ... التصميم العمراني والمباني والطرق والممرات

What are the different forms of virtual games?

- **Commercial PC**– This type of video games are played on personal computers. They are more diverse and include user-determined hardware and software. They generally have higher capacity in input, video and audio output processing.
- **Console Games** – Console games are a form of interactive multimedia entertainment. They comprise manipulated images and sound generated by the gaming console and can display them on the Television or a similar audio-video system. The game is usually managed and controlled with a handheld device called controller, connected to the console system.
- **Indie Games** – Indie games are independent video games, and they are most often created without the publisher's financial support. These are innovative games and mostly rely on digital distribution.
- **Mobile Games** – Mobile games are played on feature phones, smartphones, tablets, PDA, smartwatch, graphics calculator, and a portable media player. These games are usually downloaded from the app stores and the portals of the mobile operators.



Ref : <https://www.goodfirms.co/blog/best-free-open-source-game-development-software-solutions>

Video Game Industry Statistics In 2020

1. PC Gaming
2. Console Gaming
3. Mobile Gaming
4. Online gaming
5. Gaming Video Content

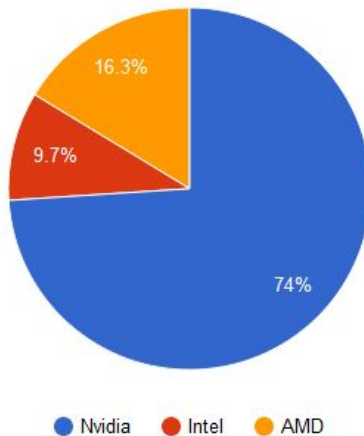
<https://www.wepc.com/news/video-game-statistics/>

Video Game Industry Statistics In 2020

1-PC gaming

one of the most popular forms of video gaming (despite increasingly impressive competition from both the console market and mobile gaming platforms)

PC Video Card Usage by Manufacturer



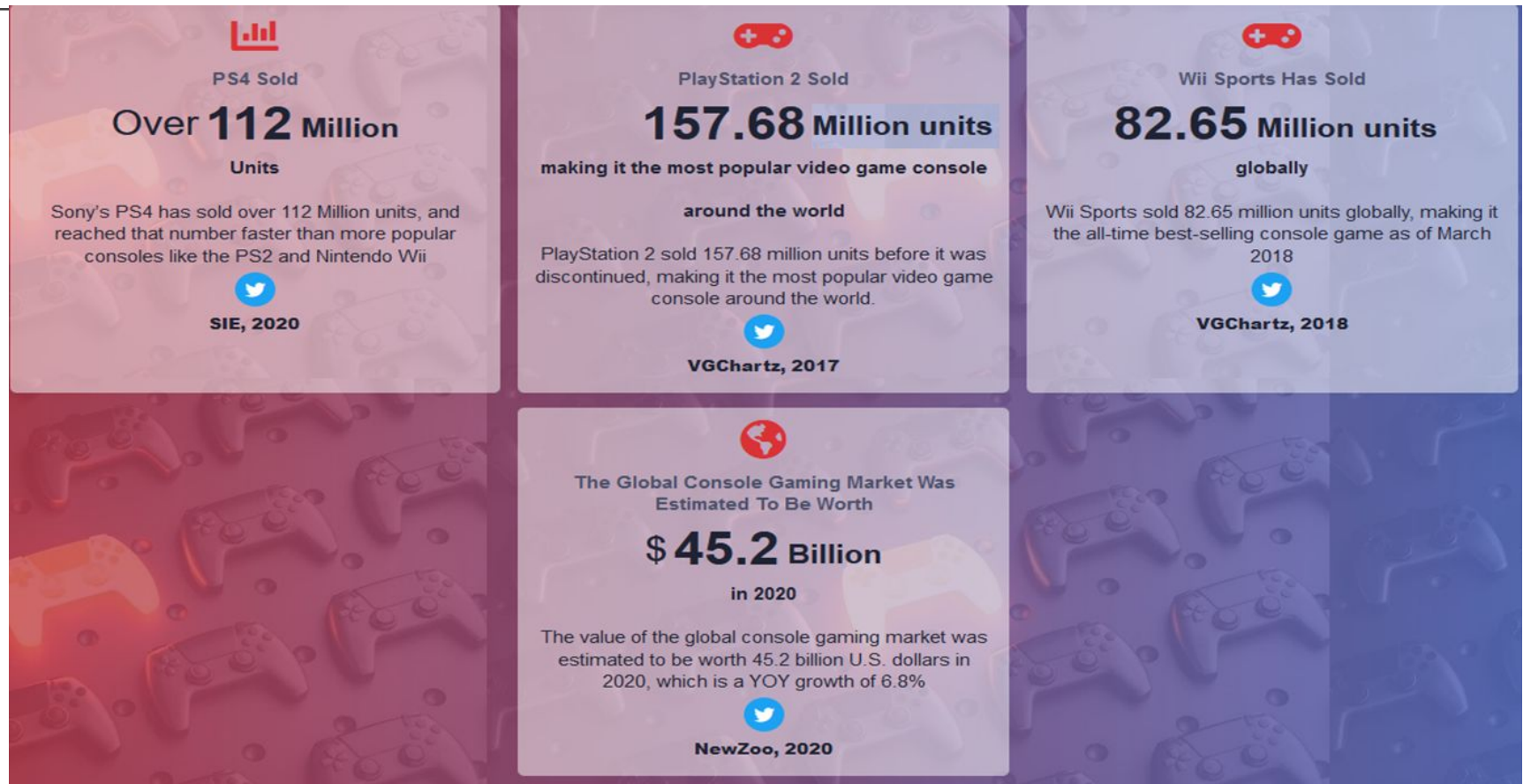
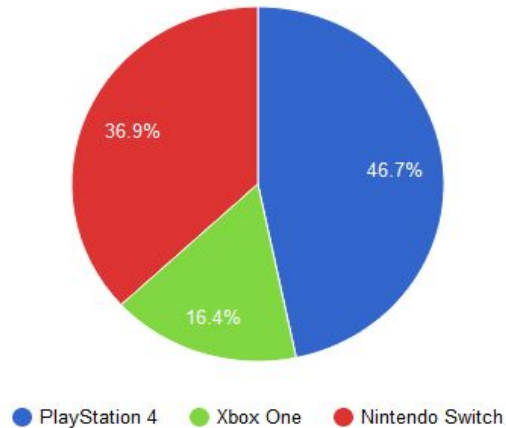
<https://www.wepc.com/news/video-game-statistics/>

Video Game Industry Statistics In 2020

2-Console Gaming

At the time of writing, the **PlayStation 2** is still the best-selling console of all time with 157.68 million units sold, and the best-selling console game is still **Wii Sports** with 82.65 million units.

Current-gen market share by platform



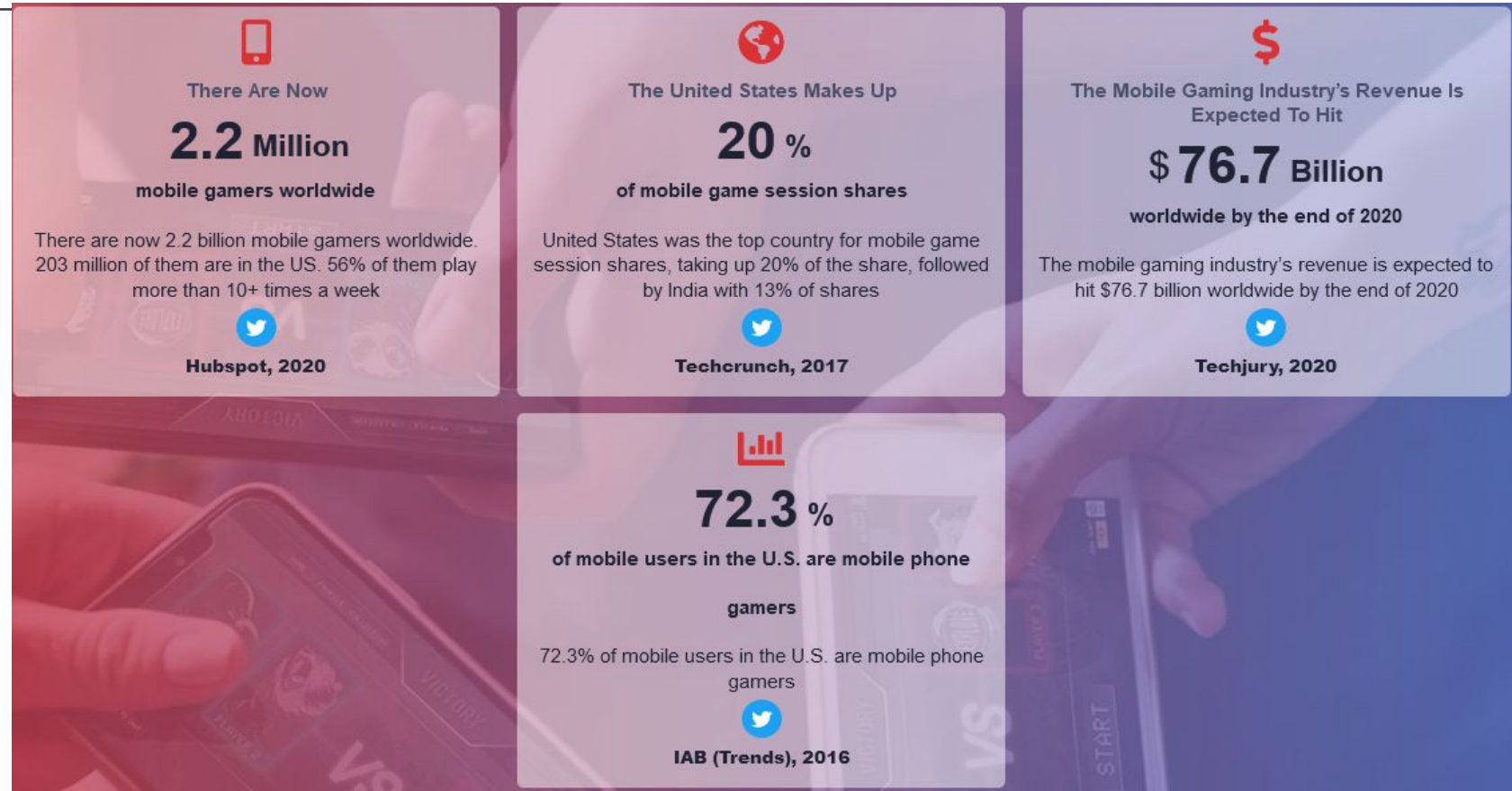
<https://www.wepc.com/news/video-game-statistics/>

Video Game Industry Statistics In 2020

3- Mobile Gaming

Mobile gaming has fast become the largest gaming market in the world with industry revenue expected to hit \$76.7 Billion by the end of 2020 and 2.2 million mobile gamers worldwide.

72.3% of mobile users in the US are mobile phone gamers.



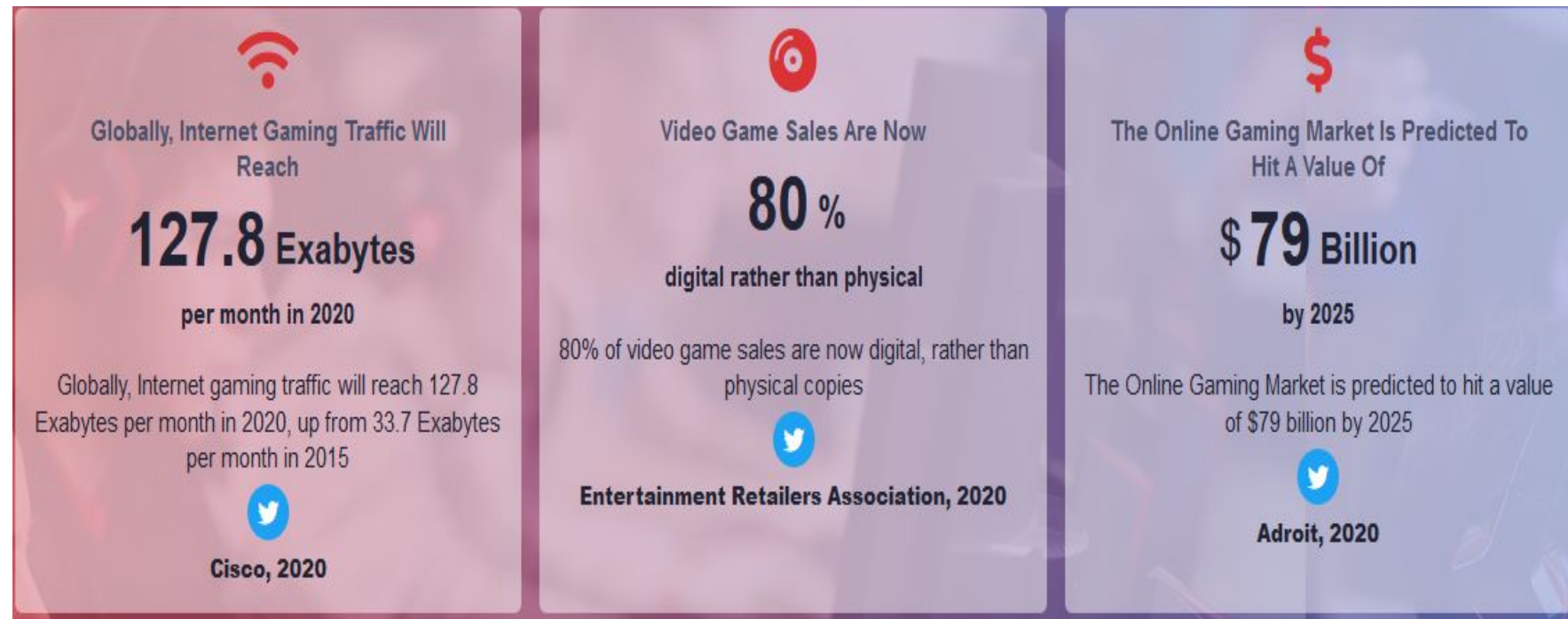
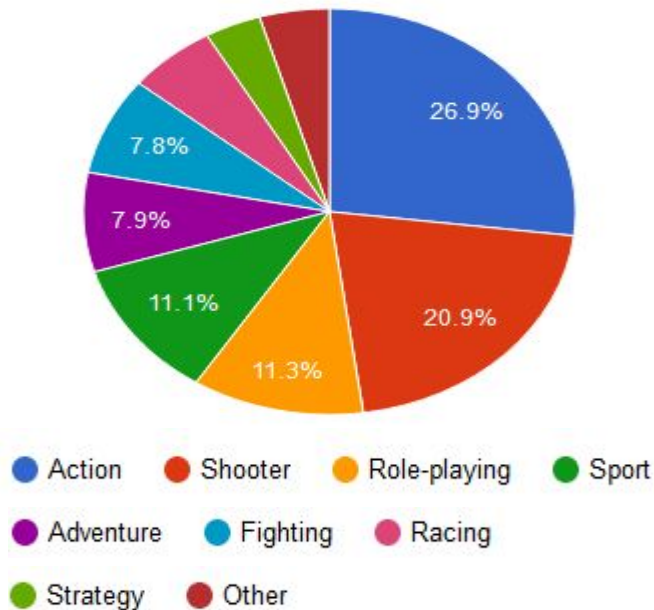
<https://www.wepc.com/news/video-game-statistics/>

Video Game Industry Statistics In 2020

4-Online Gaming

The online gaming market is predicted to reach a value of \$79 billion by 2025 as more and more people move to game online.

Genre breakdown of video game sales in the US (2018)



Note: an **Exabyte** = 1000⁶ byte (Wikipedia)
<https://www.wepc.com/news/video-game-statistics/>

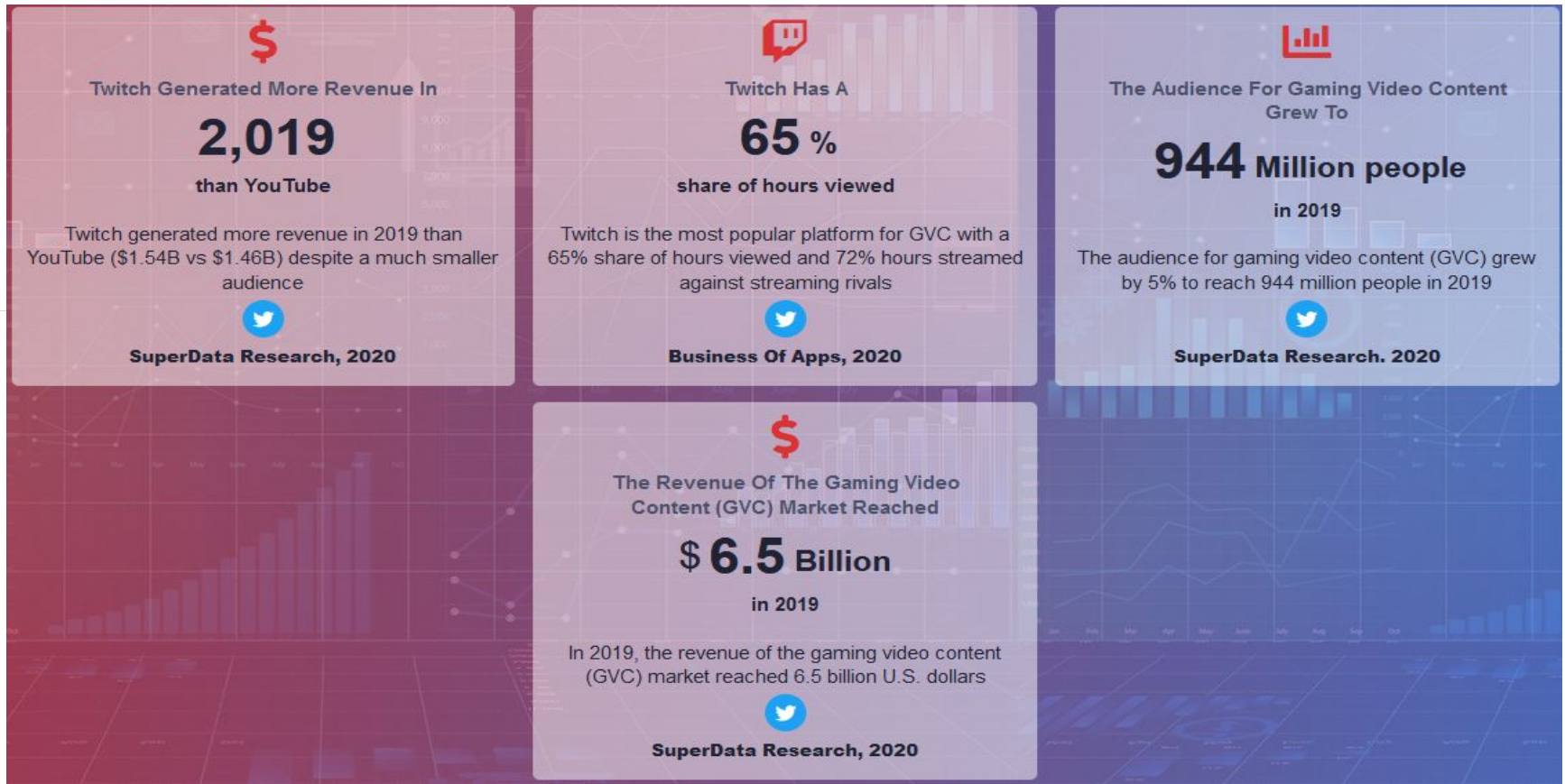
Video Game Industry Statistics In 2020

5- Gaming Video Content

The Gaming video content market has become incredibly popular in the last decade with platforms like Twitch and YouTube capitalizing on the rise of both video games and video streaming. In fact, the revenue for the Gaming video content (GVC) market reached \$6.5 billion in 2019 making it an incredibly lucrative industry.

Note: **Twitch** is a video live streaming service that focuses on games live streams.

Average Concurrent Viewers on Twitch



<https://www.wepc.com/news/video-game-statistics/>

Famous Game Genres

- Shooter Games
- Action and Arcade Games
- Strategy Games
- Role-Playing Games
- Sports Games
- Vehicle Simulations
- Construction and Simulation Games
- Adventure Games
- Puzzle Games
- Serious Games

Serious Games [educational]

(wikipedia) A **serious game** or **applied game** is a game designed for a primary purpose other than pure entertainment.

used by industries like education, scientific exploration, health care, emergency management, city planning, engineering, and politics

Example: SimSE game for learning Software Engineering models
(waterfall, xp,)

<https://www.ics.uci.edu/~emilyo/SimSE/downloads.html>

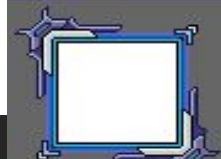


ARTIFACTS
CUSTOMERS
EMPLOYEES
PROJECTS
TOOLS



Current Activities:

- Andre
- Anita
- Calvin
- Emily
- Mimi
- Pedro
- Roger



TIME ELAPSED

00000000

TICKS

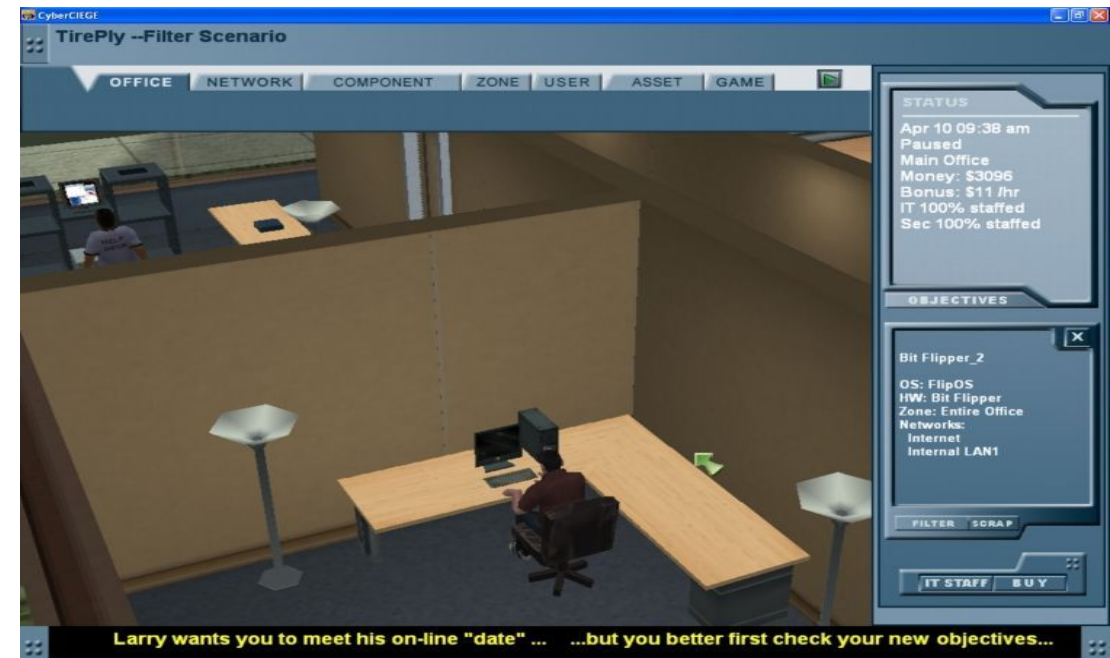
1

NEXT EVENT
ADVANCE CLOCK
STOP AT EVENTS

CyberCIEGE

2005 – Naval Postgraduate School and
Rivermind, Inc.

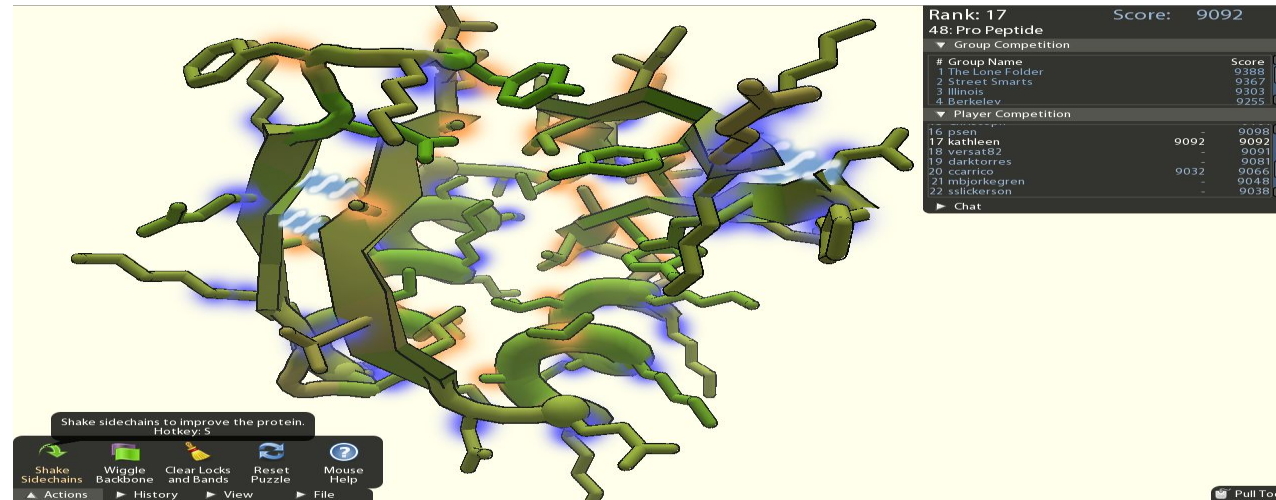
CyberCIEGE is a serious game designed to teach network security concepts. Its development was sponsored by the U.S. Navy, and it is used as a training tool by agencies of the U.S. government, universities and community colleges. The game is freely available to the U.S. Government and a no-cost license is available for educational institutions.



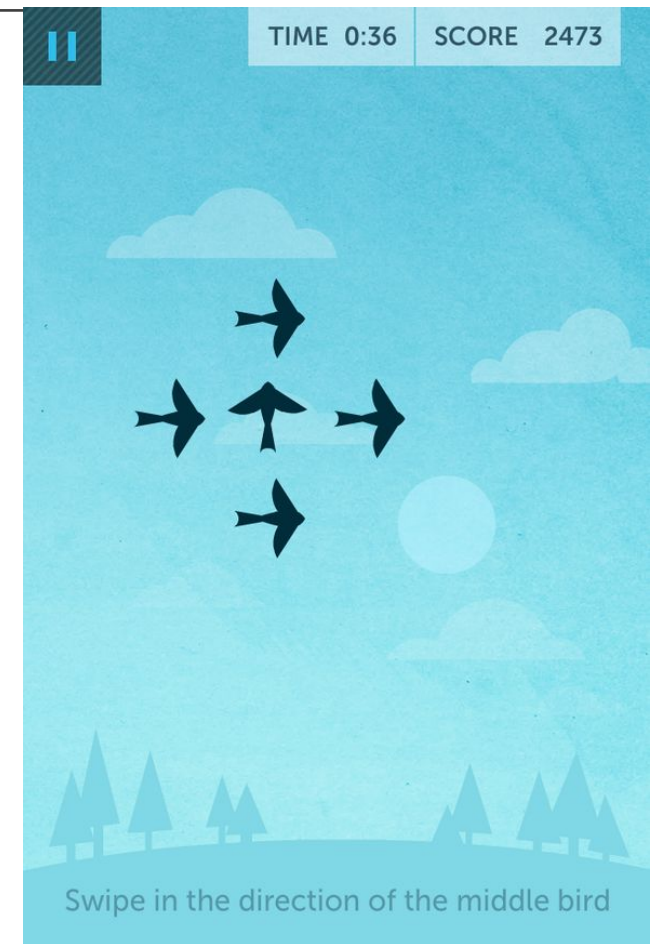
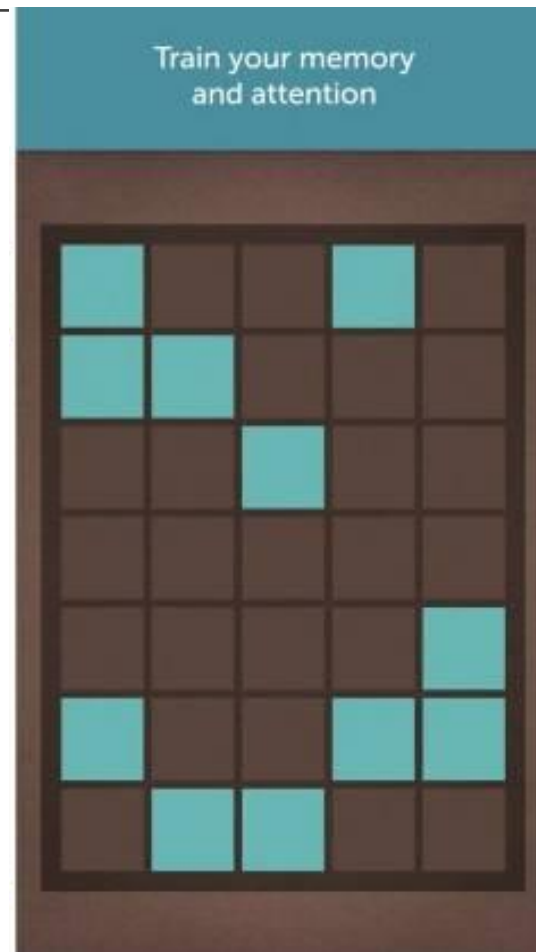
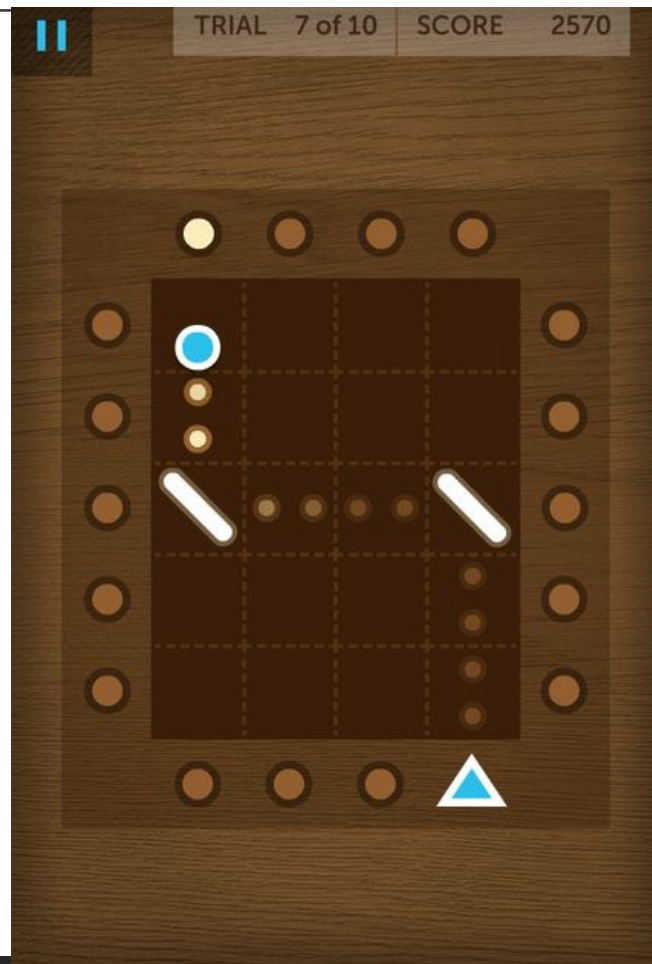
Foldit

2008 – David Baker (Univ of Washington)

Foldit is an online puzzle video game about protein folding. The game is part of an experimental research project, and is developed by the University of Washington's Center for Game Science in collaboration with the UW Department of Biochemistry. The objective of the game is to fold the structure of selected proteins to the best of the player's ability, using various tools provided within the game. The highest scoringsolutions are analysed by researchers, who determine whether or not there is a native structural configuration (or native state) that can be applied to the relevant proteins, in the "real world". Scientists can then use such solutions to solve "real-world" problems, by targeting and eradicating diseases, and creating biological innovations.



Lumosity.com

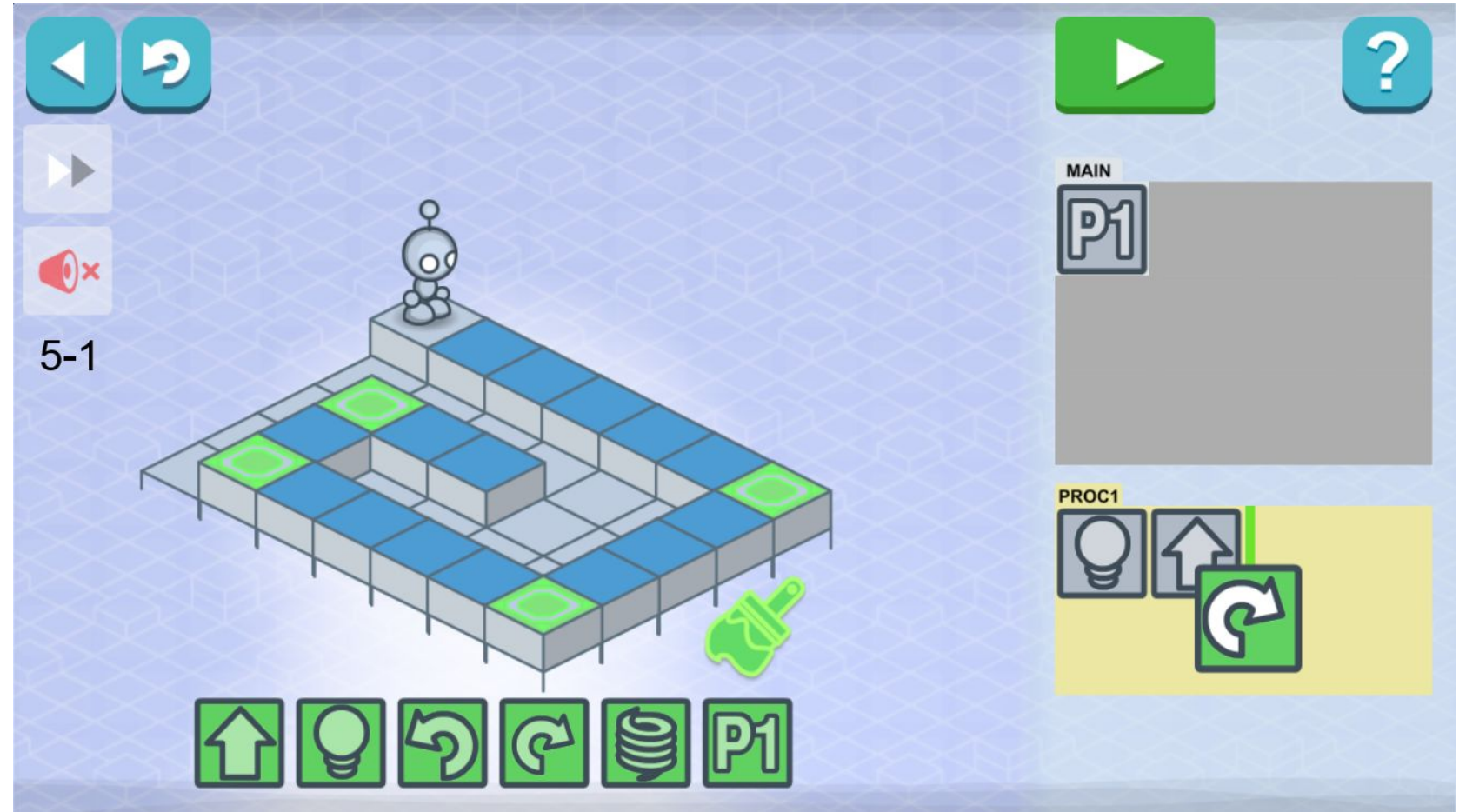


Power Up



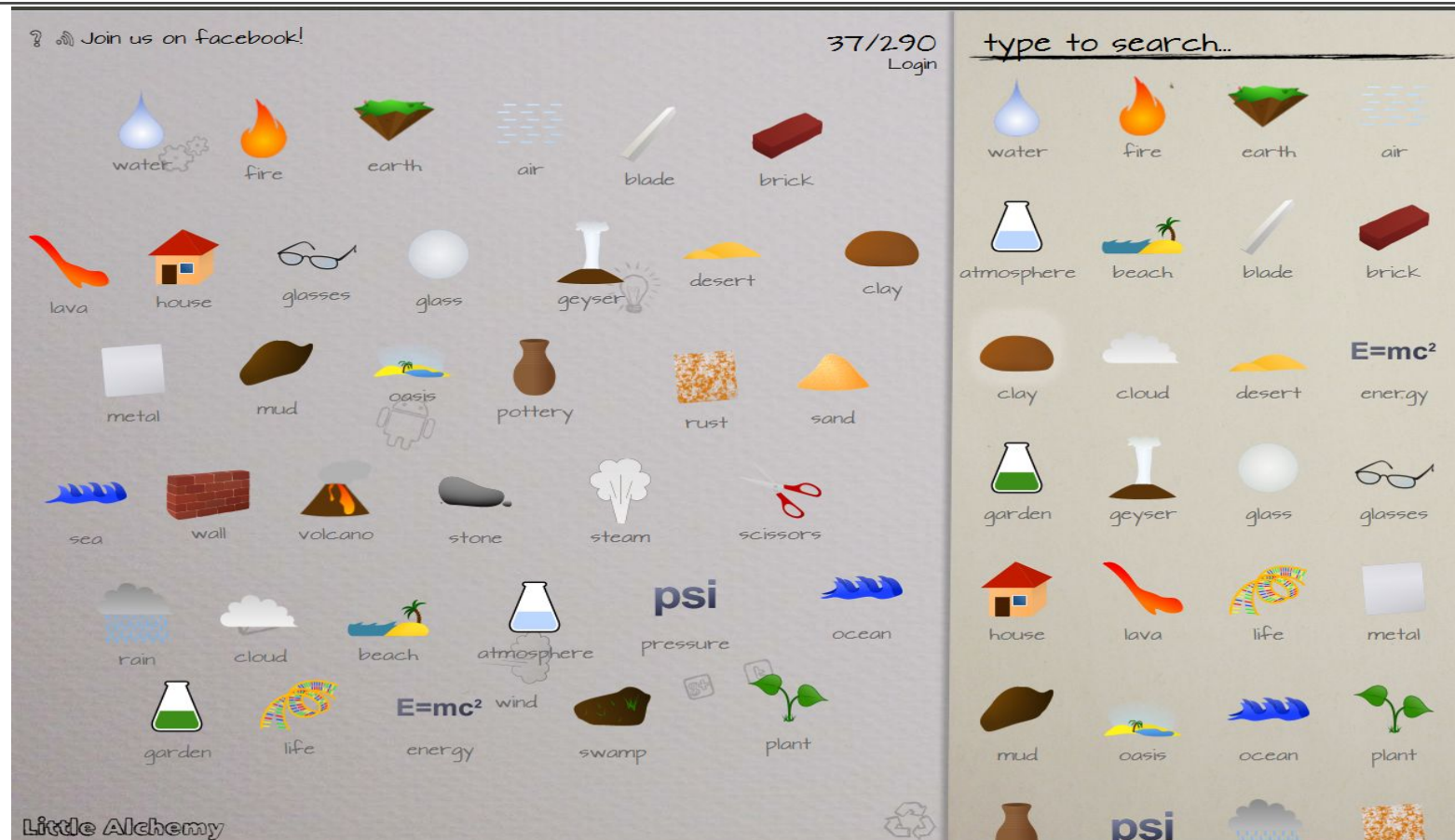
Lightbot

Lightbot has proven popular with kids who are learning STEAM (Science, Technology, Engineering, Art, and Maths) skills. On Lightbot's website and in its apps, children learn coding by solving numerous puzzles. There are two levels of the game – one for ages 4-8 and the other for ages 9 and up. As coding becomes a mandatory skill for today's youth, Lightbot players are already at a huge advantage.

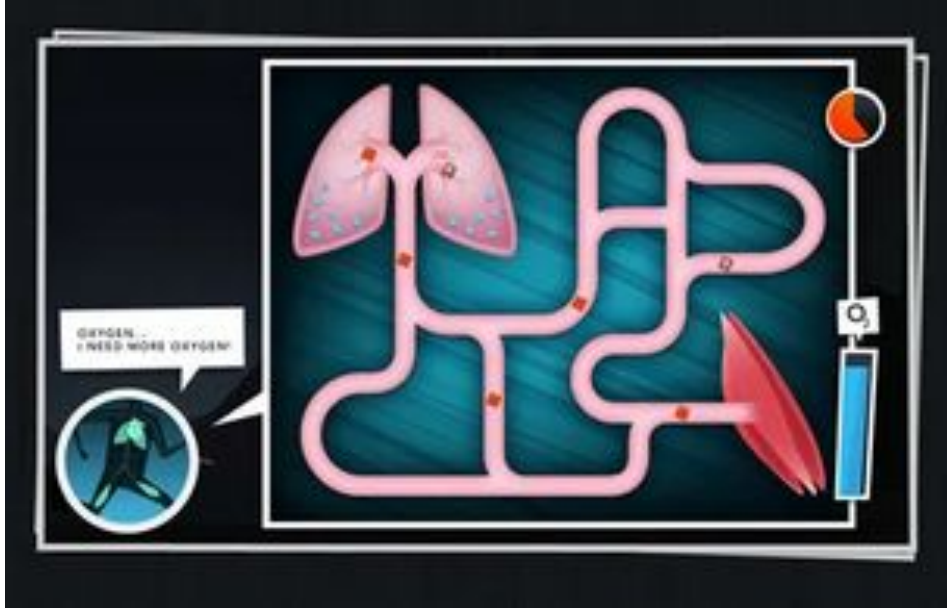


Little Alchemy -

[/https://littlealchemy.com](https://littlealchemy.com)



Code Fred



رابط اللعبة

<http://www.msichicago.org/play/codefred>

الهدف:

تهدف اللعبة للتعرف على العمليات الحيوية التي تتم داخل جسم الانسان في لحظات الحركة، فكرة اللعبة بسيطة: هناك شخص يطارده وحش، ولا بد من الهروب منه بأقصى سرعة، وبالتالي مطلوب منك أن تعرف كيف تزيد معدل نبضات القلب، كيف توصل الاكسيجين للعضلات

LEGO® Juniors Create & Cruise



2.Existing business models that capitalize on this technology with description of sample commercial products in each business model

Types of Gaming Business models :

1. **Game purchase (Premium games):** where the digital gaming company sells it at a given price.
2. **In-app purchases (Freemium games):** similar to the free-to-play model, the free game prompts gamers to buy either digital goods or premium versions/upgrades of the game.
3. **Ad-supported:** another model is the ad-supported, where gamers get it for free, and advertising is rendered within the game, which will monetize on both an impression and click basis.
4. **Hybrid model games:** A mix between In-app purchases and Ad-display which lets you pay to get rid of the advertisements.
5. **Crowdsourcing model:** You crowdsource (collect) the data/information from users and sell it to someone who needs the information. The game is designed in such a way that the users don't even mind doing the work for you (Privacy Concerns).
6. **Licensing :** There are more and more portals interested in showing your games on their websites. They follow various strategies to earn money via your games, but you don't have to worry about all that as selling the license is usually a one-time deal.
7. **non-game focused monetization**

<https://fourweekmba.com/gaming-industry/>,

<https://www.feedough.com/how-to-choose-the-right-business-model-for-your-game/> ,

https://developer.mozilla.org/en-US/docs/Games/Publishing_games/Game_monetizati

Tools

- a. Common tools (Commercial)
 - i. Illustrator
 - ii. Photoshop
 - iii. Unity 3D
 - iv. Maya /3D max
 - v. Maximo
 - vi. Photon PUN2
 - vii. Other
- b. Common Art sites for Indie Programmers

Most popular Game Engines

1. Unity Game Engine
2. Unreal Engine
3. Godot
4. GameMakerStudio
5. Phaser 2D

Comparison of Game Engines 2020



Ref : <https://indiegamedev.net/2020/02/11/comparison-of-game-engines-2020/>



THE TOP 10 FREE GAME DEVELOPMENT TOOLS



TOOLS



1

GameMaker: Studio

<http://www.yoyogames.com/studio>

Game Maker is easily one of the most popular game development tools in the industry. This is because its drag-and-drop system allows you to make games many times faster than if you were coding it.

You also have the option of upgrading to the Professional or Master Collection versions for access to more tools.

3

Stencyl

<http://www.stencyl.com>

With this toolset, you can make Flash games faster than with any other program out there. The interface features a drag-and-drop system along with blocks of code for you to utilize.

While you can make great games with Stencyl, it's often tool by developers to make quick prototypes of their game ideas.

5

Cocos2D

<http://www.cocos2d.org>

Cocos2D is an open-source tool for creating 2D games with C++, although there is support for Lua and JavaScript.

Your finished game can be published on Windows, Mac, Android, iOS, Windows Phone, or the web. Notable titles made using this program include Badlands, Castle Clash, Kingdom Rush, Tiny Village, and Matching with friends.

2

Unity

<http://unity3d.com>

For a long time, many developers underestimated Unity, but not anymore. It is now a widely-used game creation tool that can be used to make both 2D and 3D games on just about any popular platform.

There is also free publishing support for iOS, and the free download comes with a trial of Unity Pro.

4

Construct 2

<https://www.scirra.com>

Construct 2 is the perfect tool if you want to make great 2D games. This game creation engine offers an intuitive drag-and-drop interface that also lets you make use of your HTML5 knowledge.

This means you can make games for NewGrounds, Facebook, Kongregate, and popular web browsers.

6

PlayCanvas

<http://www.playcanvas.com>

This HTML5 game development platform allows you to develop 2D and 3D games alongside a collaborator on the cloud.

This means you don't have to download and install anything and are even given 200MB of disk space. New developers and experts alike can make use of this free and open-source engine.

7

RPG Maker

<http://www.rpgmakerweb.com>

RPGs are one of the most popular genres in the industry. Thanks to RPG Maker, thousands have had the chance to make their epic adventures with this excellent game development engine.

From combat, cutscenes, and dialogue to environments, dungeons, and bosses, you can do almost anything without having to write a single line of code.

9

Unreal Engine 4

<https://www.unrealengine.com/what-is-unreal-engine-4>

In March of 2015 Epic Games announced that Unreal Engine 4 would be free to download and use for everyone. This renowned game development toolset has been used for hits on just about any platform.

They even only ask for 5% royalty, which is amazing considering all the powerful tools and features you're getting.

8

Marmalade

<https://www.madewithmarmalade.com>

This powerful tool lets you make anything from single-player 2D games to multiplayer 3D titles. While it does require C++ knowledge, Marmalade SDK is the quickest way to create cross-platform C++ games for several platforms.

Successful titles like Plants vs Zombies and Godus were made with Marmalade.

10

MonoGame

<http://www.monogame.net>

MonoGame lets you use your knowledge of .NET languages and C# to create games for iOS, Android, PlayStation 4, macOS, and more.

More than one thousand games have shipped thanks to a healthy community of developers helping each other understand and get better with MonoGame.

BONUS

GameSalad

<https://gamesalad.com>

Thousands of games have been built with GameSalad for iOS, Android-based devices, and more.

This is because it offers a web-based interface that allows you to release a game on multiple platforms. Many games developed with GameSalad have reached the top 100 games in the US App Store and even a few in the #1 spot.



Game Revenues

1. Freelancing
2. Selling your own games

10 Best Freelance Websites

1. [Toptal](#)
2. [Upwork](#)
3. [Freelancer](#)
4. [Fiverr](#)
5. [Guru](#)
6. [PeoplePerHour](#)
7. [iFreelance](#)
8. [99Designs](#)
9. [SimplyHired](#)
10. [Craigslist](#)

مواقع عربية للعمل عن بعد

1. SouqFann سوق فن -
2. Mostaql.com مستقلّ -
3. Ureed أريد -
4. tasmeeemME تصميمي -
5. Khamsat خمسات -

<https://www.for9a.com/learn/%D9%85%D8%A7-%D9%87%D9%8A-%D8%A7%D9%81%D8%B6%D9%84-10-%D9%85%D9%88%D8%A7%D9%82%D8%B9-%D9%84%D9%84%D8%B9%D9%85%D9%84-%D8%A7%D9%84%D8%AD%D8%B1-%D8%A7%D9%84%D8%B9%D9%85%D9%84-%D8%B9%D9%86-%D8%A8%D8%B9%D8%AF>

3 ways to sell your indie games

1. Sell via platforms for gamers

Some of the most popular distribution platforms include Steam, Desura, and GOG. IndieCity, IndieVania, Green Man Gaming, WildTangent Games and GameStreamer are somewhat less recognizable but they still reach thousands of players globally.

Revenue split

Digital distribution platforms will often have their share in your profits. Unfortunately, their revenue split hovers around **50%** of a game's price.

The bigger the platform charge, the higher the price of your game, and thus it is less attractive to players. Alternatively, you can always keep the price low and resign yourself to lower earnings. Moreover, some platforms even make decisions about discounts without consulting the developer. This may also mean lower profits.

3 ways to sell your indie games

2. Sell to Game business

As opposed to gamer's distribution platforms, developers can also sell their games to wholesale and retail businesses. You can easily submit your game to the platform and offer it to business clients globally with [CodesWholesale.com](https://codeswholesale.com). Here you can upload your game activation codes and set your own price.

3 ways to sell your indie games

3. Sell via your own website

Last, but not least, you can start your own website to make a profit from an indie game. This may be your dev studio website or a website dedicated to a particular game. Online presence is essential if you want to reach a wider audience of players.

Your checklist for the App Store and Google Play

App Store	Google Play
•Unique app name •iOS 11 SDK features •Support for iPhone X Super Retina display •Icon •5+ screenshots for iPhone and 5+ for iPad •Description •Information on what's new (if submitting a new version) •Owner's name + contact information •Category •Rating •Keywords (tags) •Support and marketing URLs •Test account (for review team) •TestFlight beta testing (inviting users to test app before release)	•Unique app name •Icon (512x512 px) •2–8 screenshots •Feature graphic (1024x500 px) •Short description •Full description •Content rating •Privacy policy

Best places to publish your PC Indie Game

PC Storefronts for Indie games

1. Steam
2. Itch
3. GOG
4. Humble
5. GameJolt
6. Discord
7. Kartridge
8. Facebook Gameroom

Best places to publish your PC Indie Game

1- Steam

Steam has become a go to place for PC gaming. Millions of players visit it and play games through Steam every month. With this fact in mind standards on Steam are pretty high, meaning not every game is welcomed with opened arms from players. But, if you're publishing high quality indie game you definitely want to keep Steam high on your list of priorities.

Monetization:

- Steam lets you sell your game on their platform,
- Also, Freemium games have become quite popular on Steam as well. This basically means you can publish your game for free and add in app purchases in it (eg. more levels, skins etc.).
- Just make sure not to put ads in it, PC platform in general just isn't supportive of that.

Best places to publish your PC Indie Game

1- Steam [2]

Publishing process: Steam has recently changed it's model for publishing games on their platform. They became more open for small, indie developers by letting go of Greenlight and with the introduction of Steam Direct. With this new model publishing your game to Steam has never been easier:

1. Sign up for Steamworks <https://partner.steamgames.com>
2. Create your Steam build and deposit \$100 (this ensures that your game isn't spam and is recoupable قابلة للاسترداد after you make \$1000)
3. Implement Steamworks features into your game (e.g. Microtransactions, Matchmaking, Anti-cheat, Cloud storage, Management of user-generated content...)
4. Create your store page and setup pricing
5. (it is recommended that you set up your store **listing before even** publishing your game, this way you can get more people **wishlisting** يترقبون your game)

Best places to publish your PC Indie Game

2- Itch [١]

Itch is an open marketplace for independent creators. It is one of the more popular platforms for **indie games** on the market right now. It supports PC, web and even mobile (android) games.

Monetization:

- They let you customize your revenue share (open nature حرية في تقسيم الأرباح).
 - By default you're giving 10% of your revenue to Itch,
 - but this can be set to 0% as well.
- Itch also has a '**pay what you want**' system. This lets passionate players **donate** money to your free game, or even pay up your paid game.

Best places to publish your PC Indie Game

2- Itch [2]

Publishing process: Publishing your game on Itch is as easy as it can get:

1. Create an account on <https://itch.io/developers>
2. Create and upload new project
3. Edit your game page to your liking
4. Publish, no fees necessary

تنويه: الجزء التالي من slides للاطلاع فقط

Games ARTIFICIAL STUPIDITY

WHEN NPCS SHOULD MAKE MISTAKES

Intelligence != Fun

What makes a game entertaining and fun does not necessarily correspond to making characters smarter

The player is, after all, supposed to win

11 Ways to let NPC (non-playable character) seems to be stupid

1. Don't Cheat

Sounds easy, but many games “cheat” by making NPCs omniscient

- Know where enemies are even without seeing them
- Know where to find weapons or ammo

Players usually eventually detect cheating or at least get the feeling that the NPC's behavior seems somehow “unnatural”

2. Always miss the player the first time

It's not fun to suddenly and unexpectedly take damage

Player may feel cheated, particularly if attacked with a weapon that kills the player or does a lot of damage

By missing the player the first time, it gives the player a second to react and still keeps the tension high

3. Have horrible aim [wide cone]

Having abundant gun fire in the air keeps the player on the move and the tension high

However, the player is supposed to win

By giving NPC bad aim, one can have abundant gun fire without being too hard on the player

“Half-Life” used a wide spread on NPC weapons (as much as 40 degrees)

4. Never shoot when first see the player

When a player first walks into an area and is spotted by an enemy, the enemy should never attack right away

A secondary activity, such as running for cover or finding a good shooting location is more desirable

Gives player time to react

s. Warn the Player

Before attacking the player, warn the player that you are about to do so

- Make a sound (beep/click)
- Play a quick animation
- Say “Gotcha!”, “Take this”

This is particularly important when attacking from behind

6. Attack “kung-fu” style

Player is usually playing the role of “Rambo” (i.e. one man taking on an army)

Although many NPCs may be in a position to attack the player, only a couple should do so at a time

The remaining NPCs should look busy, reloading, changing positions, etc.

7. Tell the player what you are doing

Interpreting the actions of NPCs can often be subtle

Complex behaviors are often missed by the player. (Lot's of work for nothing)

NPCs should tell the player what they are doing

- “flanking!” “cover me!” “retreat!”

Players with often intuit intelligence behavior that isn't really there

g. Intentionally be vulnerable

Players learn to capitalize on opponent's weaknesses.

Rather than allowing the player to discover unintentional weaknesses in the AI, vulnerability should be designed into an NPC's behavior.

- Stop moving before attacking
- Pause and prepare weapon before attacking
- Act surprised and slow to react when attacked from behind

Planned vulnerability makes the characters seem more realistic

Unintentional mistakes break the realism (seems like fighting a computer program)

9. Don't be perfect

Human players make mistakes

When NPCs behave perfectly they seem unnatural

If an NPC knows how to avoid trip mines, run into them occasionally

When reloading, sometimes fumble with the gun

10. Pull back last minute

Trick:

Push the player to the limit

Attack vigorously until the player is near death

Then pull back. Enemy becomes easier to kill

Makes player feel like they really accomplished something

11. React To Mistakes

Mistakes in AI are inevitable

Unhandled, they make make the AI look dumb

By recognizing mistakes and reacting to them intelligently they can be turned into features

11. React To Mistakes

Example 1:

- Occasionally when an NPC throws a grenade, it bounces off of another object and lands back at the NPC's feet
 - (Note that the player occasionally makes this mistake too!)
- Looks dumb as the NPC blows himself up
- If the NPC reacts, however, the mistake turns into a feature:
 - NPC body and facial expression can show surprise, fear
 - NPC can say "Oh Shoot!" or "Doh!"

Thanks

